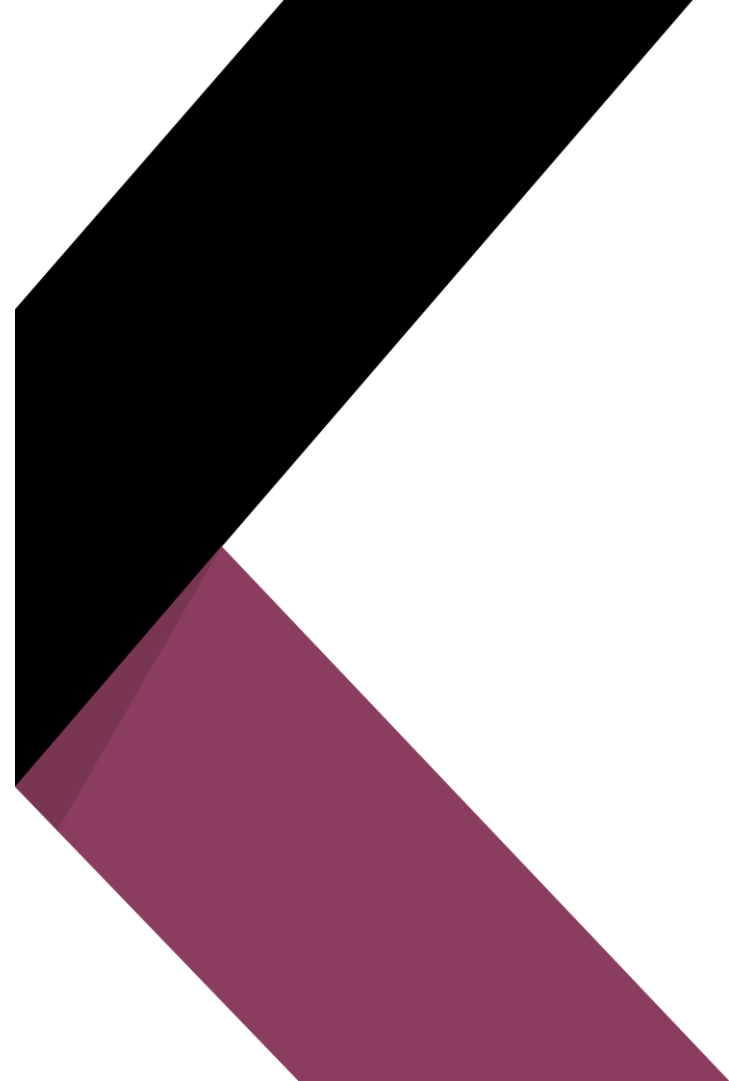


Test Case Creation





Test Case Creation

Creating a test case is a fairly laborious process. The first thing to do is think in terms of customer and try to understand, always starting from a VF or a diagnostic document, which could be the various test cases compared to a specific feature.

In this regard we can distinguish two macro categories of test cases:

- functional test cases
- diagnostic test case



Functional Test Case

Functional Tests represent the first step in guaranteeing software quality, because they allow us to verify not only that the application functions as expected on the technological front but, above all, that it does what users actually expect with respect to their initial requests.

Functional tests, in fact, take into account the needs of users, give developers the certainty that business processes meet these needs and allow Qa (Quality Assurance) teams to verify that the software is actually ready for release in production.



Functional Test Case - Where to start -1-

To get to the writing of a functional test case we start from the analysis of a VF, from which we extract all the possible cases of operation, converting everything into test case.

Possible test cases can be extracted from descriptive text, from a descriptive table, from a truth table, from a logical equation, from a state machine, from a flowchart.



From descriptive text

4.1.4.2 Wake up Initiated by the Master

If the Master is awakened by a Hardware Wake up that requires the CAN C bus to be awake, it shall transmit the WAKE_C_NBC message, setting the MainWakeSts bit and the NodeX bit corresponding to itself.

Similarly, the Master shall also send its WAKE_C message with active MainWakeSts and NodeX bits, when it recognizes that the CAN BH bus is active. Figure 2 below shows a skeleton diagram of the Class B Master State Machine as defined in TFO 07320, Figure 2. The diagram below is a reference to show the ADDITIONAL activities that the Master must perform to coordinate waking the CAN C busses when the CAN BH bus is awake. (Note: putting both the CAN C busses and the CAN BH bus to sleep will be addressed in state chart describing the CAN C Master later in this document.)

<Picture>



Functional Test Case - Where to start -3-

In case of missing message on CAN the NAG/NAP behaviour on LIN as follow:

From a descriptive table

Message State	LIN signals	
	InternalLightSts	InternalLightLevel
	Status	Level
NBC received NQS received	CAN value	CAN value
NBC received NQS missing	Not Active	100%
NBC missing NQS received	Active	CAN value
NBC missing NQS missing	Active	100%



Functional Test Case - Where to start -4-

1.4.2 Remarks

The condition of “+ luci” is transmitted on the CAN network (*InternalLightSts* signal). The state of this signal is determined according to the following table:

<u>PosLightCmd (+ luci)</u>	<u>NightDaySts</u>	<u>InternalLightSts CAN message</u>
0 (Position lights OFF)	0 (night condition)	0
1 (Position lights ON)	0	1
0	1 (daytime condition)	0
1	1	0

From a truth table



Functional Test Case - Where to start -5-

From a logical equation

"InternalLightSts" = (PosLightCmd OR RearPositionNightCmd) AND NOT(NightDaySts).



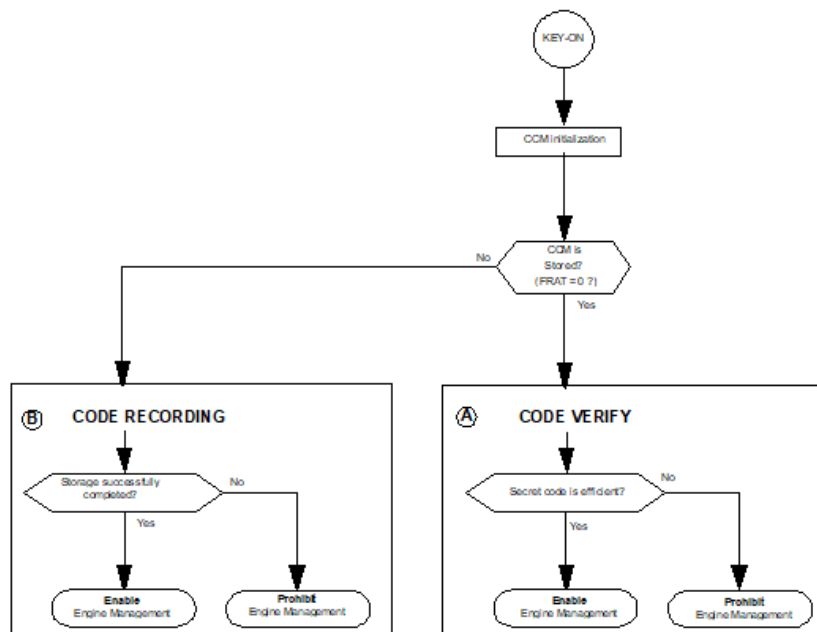
From a state machine





Functional Test Case - Where to start -7-

From a flowchart





Functional Test Case - Where to start -8-

In all cases, except for the case of the descriptive text, the process is simpler and more straightforward, because what is written must be followed and there is not much margin of imagination to be able to extract some tests. This is because you can not change a truth table or a flowchart, so just follow what is written and put it in the form of tests, using the syntax and form chosen for writing the tests themselves.



Diagnostic Test Case

The rules relating to the functional safety of electronically managed machines and systems identify a particular type of test; diagnostic tests, which are intended to detect any faults in the electronic components. The timely detection of faults, especially if they are dangerous, is essential to keep the system or the machine in a safe state, even in the face of serious anomalies; in fact, the level of safety integrity (SIL) strongly depends on the degree of diagnostic coverage and the fraction of dangerous failures detected.

In this regard, diagnostic test cases are useful to search for malfunctions ("fault") when these happen and take actions to minimize damage first that the actual failure occurs ("failure").



Diagnostic Test Case - Where to start -1-

To get to the writing of a diagnostic test case we start from the analysis of a VF, from which we extract all the possible diagnosis cases, converting everything into test case.

Possible test cases can be extracted from dedicated section in the VF (if present) and from more detailed diagnostic documents called CDD.



Diagnostic Test Case - Where to start -2-

In a VF, for example, a section dedicated to diagnosis can be found (if present), which can be structured as follows:

1.10.4.1 ID 23.0 Description

IF, after a T_Code_Missing_Msg_NBC_NPE time, the NBC doesn't receive the message TRM_CODE_RESPONSE
THEN

the NBC shall:

1) set a specific DTC according to the "NBC Diagnostic Requirement" document.

WHEN the NBC receives correctly the message TRM_CODE_RESPONSE

THEN

the NBC shall:

1) delete the DTC previously set according to the "NBC Diagnostic Requirement" document.



Diagnostic Test Case - Where to start -3-

To check all the validation / invalidation conditions of the DTCs and the relative timescales, consult the document called CDD.

Starting from this document it is possible to develop the diagnostic test cases necessary to cover the whole part of diagnosis related to a function.

CONTACTS

Bruno D'Angelo

bruno.dangelo@kineton.it

Via G. Porzio,
Centro Direzionale Isola B3
80143 Napoli Via G. Porzio

www.kineton.it

